

TSA6213 SYSTEM ADMINISTRATION & MAINTENANCE

LAB 4

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Section	SAM1, 1A

DNS and DNS Tools

1. What is MYNIC and how it relates to DNS?

MYNIC is the official .MY domain registry recognised by ICANN. MYNIC provide Malaysian organisations and individuals their unique brand identity on the Internet. It relates to DNS as it maintains authoritative DNS records for domains under .MY, ensuring their proper resolution to IP addresses.

2. Use MYNIC whois service and find information about MMU DNS information. What are MMU Primary Server, Secondary Server and IP address?



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Domain Name	mmu.edu.my
Domain Registration No.	D5A000024-MYNIC
Domain Registration Date	24 June 1999 00:00:00 (Malaysia Time) 23 June 1999 16:00:00 (UTC)
Domain Expiry Date	24 June 2025 00:00:00 (Malaysia Time) ⓘ 23 June 2025 16:00:00 (UTC)
Domain Last Updated	16 June 2024 16:18:45 (Malaysia Time) 16 June 2024 08:18:45 (UTC)
Domain Status	active ⓘ
DNSSEC	Unsigned Delegation
Registrar	MYNIC Berhad
Registrar Phone Number	
Nameservers	ext-ns-1.mmu.edu.my 203.106.62.231 ext-ns-2.mmu.edu.my 203.106.61.231 ext-ns-3.mmu.edu.my 203.106.60.231 ext-ns-4.mmu.edu.my 203.106.63.231

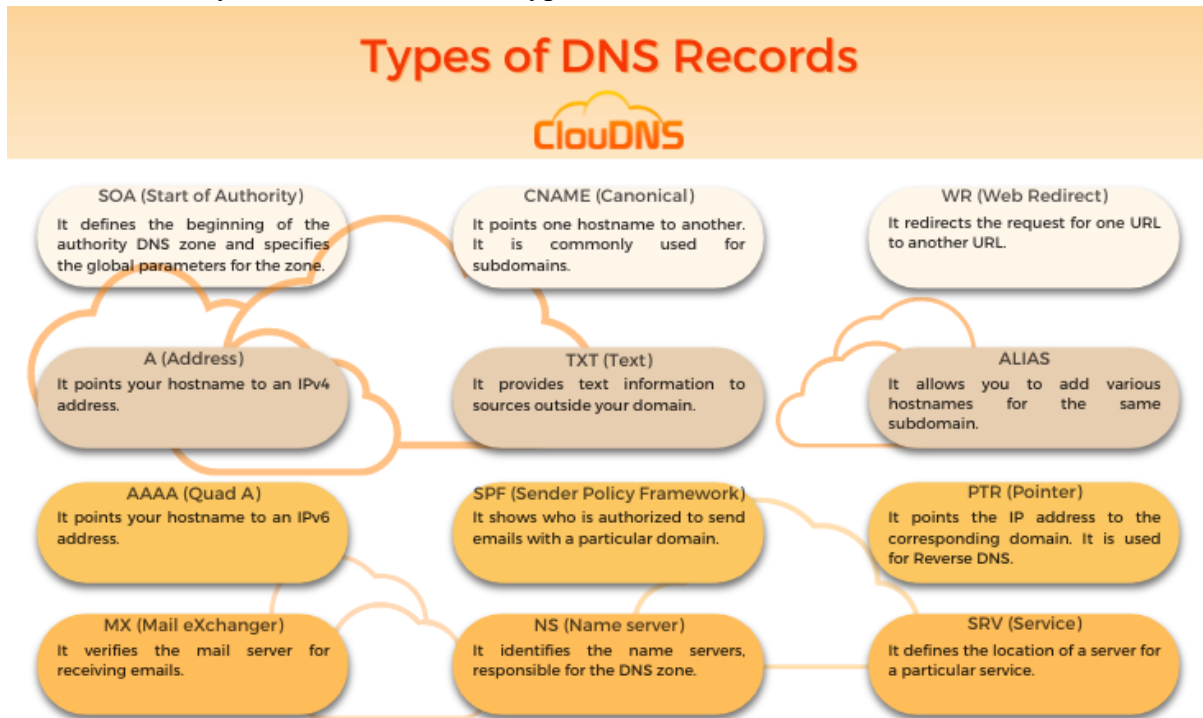
If whois have not been installed, use command `sudo apt install whois`
`whois mmu.edu.my`

```
osboxes@osboxes:~$ whois mmu.edu.my
Domain Name: mmu.edu.my
Registry Domain ID: D5A000024-MYNIC
Registrar WHOIS Server: whois.mynic.my
Registrar URL: https://whois.mynic.my
Updated Date: 2024-06-16T08:18:45.005Z
Creation Date: 1999-06-23T16:00:00.000Z
Registry Expiry Date: 2025-06-23T16:00:00.000Z
Registrar: MYNIC Berhad
Registrar IANA ID: 999000002
Registrar Abuse Contact Email: abuse@mynic.my
Registrar Abuse Contact Phone:
Domain Status: ok https://icann.org/epp#ok
Registry Registrant ID: REDACTED FOR PRIVACY
Registrant Name: REDACTED FOR PRIVACY
Registrant Organization: REDACTED FOR PRIVACY
Registrant Street: REDACTED FOR PRIVACY
Registrant City: REDACTED FOR PRIVACY
Registrant State/Province: REDACTED FOR PRIVACY
Registrant Postal Code: REDACTED FOR PRIVACY
Registrant Country: REDACTED FOR PRIVACY
Registrant Phone: REDACTED FOR PRIVACY
Registrant Fax: REDACTED FOR PRIVACY
Registrant Email: Please query the RDDS service of the Registrar of Record identified in this output for information on how to contact the Registrant, Admin, or Tech contact of the queried domain name.
Registry Admin ID: REDACTED FOR PRIVACY
Admin Name: REDACTED FOR PRIVACY
Admin Organization: REDACTED FOR PRIVACY
```

```
osboxes@osboxes:~$ nslookup mmu.edu.my
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   mmu.edu.my
Address: 45.60.15.220
```

3. List and briefly describe DNS record types.



<https://www.cloudns.net/blog/dns-records-different-types/>

SOA (Start of Authority): Shows the start of the authority DNS zone and specifies the global parameters of the zone.

A (Address): Points your hostname to an IPv4 address.

AAAA (Quad A): Points your hostname to an IPv6 address.

MX (Mail eXchanger): Verifies the mail server for receiving emails.

CNAME (Canonical): Points one hostname to another. Mainly for subdomains.

TXT (Text): Provides text information to sources outside your domain.

SPF (Sender Policy Framework): Shows who is authorized to send emails with a particular domain.

NS (Name server): Identifies the name servers, responsible for the DNS zone.

SRV (Service): Defines the location of a server for a particular service.

WR (Web Redirect): Redirects the request for one URL to another URL.

ALIAS: Allows you to add various hostnames for the same subdomain.

RP (Responsible person): Shows who is responsible for the domain name and specifies its email address.

SSHFP (Secure Shell Fingerprint record, used for Secure Shell (SSH)): When an SSH client reconnects to a server, they check the corresponding SSHFP record. If there is a match and the server is legit, it is safe to connect to it.

PTR (Pointer): Points an IPv4/IPv6 to a domain name. Mainly used for Reverse DNS.

NAPTR (Naming Authority Pointer): For mapping servers and the users' addresses in the Session Initiation Protocol (SIP).

CAA (Certification Authority Authorization): Gives ability to the DNS domain name holder to issue certificates for their domain.

Wildcard: Match requests for non-existing domain names.

4. Identify DNS tools lookup in Windows and Linux.

Windows:

nslookup

```
C:\Users\Asus>nslookup mmu.edu.my
Server:      UnKnown
Address:     10.0.237.43

Non-authoritative answer:
Name:       mmu.edu.my
Address:    45.60.15.220
```

ipconfig /all

```
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
    Physical Address. . . . . : 00-91-9E-9C-B1-4D
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::6313:1eea:c068:da73%18(Preferred)
    IPv4 Address. . . . . : 10.193.96.234(Preferred)
    Subnet Mask . . . . . : 255.255.224.0
    Lease Obtained. . . . . : Thursday, 5 December, 2024 7:04:04 AM
    Lease Expires . . . . . : Thursday, 5 December, 2024 8:04:06 AM
    Default Gateway . . . . . : 10.193.96.1
    DHCP Server . . . . . : 1.0.0.1
    DHCPv6 IAID . . . . . : 151032222
    DHCPv6 Client DUID. . . . . : 00-01-00-01-2B-BE-37-A5-00-91-9E-9C-B1-4D
    DNS Servers . . . . . : 10.0.237.43
                           10.0.236.42
    NetBIOS over Tcpi. . . . . : Enabled
```

ping

```
C:\Users\Asus>ping mmu.edu.my

Pinging mmu.edu.my [45.60.15.220] with 32 bytes of data:
Reply from 45.60.15.220: bytes=32 time=11ms TTL=53
Reply from 45.60.15.220: bytes=32 time=10ms TTL=53
Reply from 45.60.15.220: bytes=32 time=11ms TTL=53
Reply from 45.60.15.220: bytes=32 time=10ms TTL=53

Ping statistics for 45.60.15.220:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 11ms, Average = 10ms
```

tracert

```
C:\Users\Asus>tracert mmu.edu.my

Tracing route to mmu.edu.my [45.60.15.220]
over a maximum of 30 hops:

  1    8 ms    8 ms    10 ms  10.193.96.1
  2   13 ms   14 ms    7 ms  10.176.136.56
  3    *      *      *      Request timed out.
  4   10 ms    7 ms    8 ms  115.132.156.1
  5    7 ms    7 ms    7 ms  1.9.38.89
  6    9 ms    9 ms    8 ms  10.55.41.252
  7   13 ms   14 ms   13 ms  58.26.87.50
  8   12 ms   18 ms   13 ms  45.60.15.220

Trace complete.
```

Linux:

host

```
osboxes@osboxes:~$ host mmu.edu.my
mmu.edu.my has address 45.60.15.220
mmu.edu.my mail is handled by 20 ext-mail-4.mmu.edu.my.
mmu.edu.my mail is handled by 10 ext-mail-2.mmu.edu.my.
mmu.edu.my mail is handled by 10 ext-mail-1.mmu.edu.my.
mmu.edu.my mail is handled by 20 ext-mail-3.mmu.edu.my.
osboxes@osboxes:~$
```

dig

```
osboxes@osboxes:~$ dig mmu.edu.my

; <<>> DiG 9.16.1-Ubuntu <<>> mmu.edu.my
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 38619
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;mmu.edu.my.                IN      A

;; ANSWER SECTION:
mmu.edu.my.                 445     IN      A      45.60.15.220

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53)
;; WHEN: Wed Dec 04 23:36:18 UTC 2024
;; MSG SIZE rcvd: 55
```

nslookup

```
osboxes@osboxes:~$ nslookup mmu.edu.my
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   mmu.edu.my
Address: 45.60.15.220
```

ping

```
Terminal osboxes:~$ ping mmu.edu.my
PING mmu.edu.my (45.60.15.220) 56(84) bytes of data.
64 bytes from 45.60.15.220 (45.60.15.220): icmp_seq=1 ttl=255 time=11.5 ms
64 bytes from 45.60.15.220 (45.60.15.220): icmp_seq=2 ttl=255 time=11.6 ms
64 bytes from 45.60.15.220 (45.60.15.220): icmp_seq=3 ttl=255 time=11.6 ms
64 bytes from 45.60.15.220 (45.60.15.220): icmp_seq=4 ttl=255 time=10.9 ms
64 bytes from 45.60.15.220 (45.60.15.220): icmp_seq=5 ttl=255 time=11.6 ms
64 bytes from 45.60.15.220 (45.60.15.220): icmp_seq=6 ttl=255 time=11.5 ms
64 bytes from 45.60.15.220 (45.60.15.220): icmp_seq=7 ttl=255 time=12.8 ms
64 bytes from 45.60.15.220 (45.60.15.220): icmp_seq=8 ttl=255 time=12.2 ms
64 bytes from 45.60.15.220 (45.60.15.220): icmp_seq=9 ttl=255 time=11.6 ms
64 bytes from 45.60.15.220 (45.60.15.220): icmp_seq=10 ttl=255 time=11.8 ms
64 bytes from 45.60.15.220 (45.60.15.220): icmp_seq=11 ttl=255 time=11.5 ms
64 bytes from 45.60.15.220 (45.60.15.220): icmp_seq=26 ttl=255 time=14.3 ms
64 bytes from 45.60.15.220 (45.60.15.220): icmp_seq=27 ttl=255 time=12.7 ms
^C
--- mmu.edu.my ping statistics ---
27 packets transmitted, 27 received, 0% packet loss, time 26063ms
rtt min/avg/max/mdev = 10.887/12.264/16.875/1.256 ms
```

traceroute

```
osboxes@osboxes:~$ traceroute mmu.edu.my
traceroute to mmu.edu.my (45.60.15.220), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.2)  0.311 ms  0.254 ms  0.214 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
```

until 30 hops.

5. In Windows command prompt, run suitable command to get information about IP address and DNS services.

a) Run `ipconfig /all` command

```
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : 
    Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
    Physical Address. . . . . : 00-91-9E-9C-B1-4D
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::6313:1eea:c068:da73%18(Preferred)
    IPv4 Address. . . . . : 10.193.96.234(Preferred)
    Subnet Mask . . . . . : 255.255.224.0
    Lease Obtained. . . . . : Thursday, 5 December, 2024 7:04:04 AM
    Lease Expires . . . . . : Thursday, 5 December, 2024 9:33:39 AM
    Default Gateway . . . . . : 10.193.96.1
    DHCP Server . . . . . : 1.0.0.1
    DHCPv6 IAID . . . . . : 151032222
    DHCPv6 Client DUID. . . . . : 00-01-00-01-2B-BE-37-A5-00-91-9E-9C-B1-4D
    DNS Servers . . . . . : 10.0.237.43
                           10.0.236.42
    NetBIOS over Tcpip. . . . . : Enabled
```

b) Run `nslookup` command to confirm DNS information from `ipconfig /all` command.

```
C:\Users\Asus>nslookup
Default Server: UnKnown
Address: 10.0.237.43
```

10.0.237.43

6. In Linux, host is a simple utility for performing DNS lookups. Using host, find IP address for:

a) mmu.edu.my

host mmu.edu.my

```
osboxes@osboxes:~$ host mmu.edu.my
mmu.edu.my has address 45.60.15.220
mmu.edu.my mail is handled by 20 ext-mail-4.mmu.edu.my.
mmu.edu.my mail is handled by 10 ext-mail-2.mmu.edu.my.
mmu.edu.my mail is handled by 10 ext-mail-1.mmu.edu.my.
mmu.edu.my mail is handled by 20 ext-mail-3.mmu.edu.my.
```

b) google.com

host google.com

```
osboxes@osboxes:~$ host google.com
google.com has address 172.217.25.206
google.com has IPv6 address 2404:6800:4001:80e::200e
google.com mail is handled by 10 smtp.google.com.
osboxes@osboxes:~$
```

7. Another DNS lookup tool is dig. Using dig, find IP address and other information for:

a) tm.com.my

dig tm.com.my

```
osboxes@osboxes:~$ dig tm.com.my

; <<>> DiG 9.16.1-Ubuntu <<>> tm.com.my
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 15451
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;tm.com.my.                IN      A

;; ANSWER SECTION:
tm.com.my.                 900     IN      A      45.60.245.95
tm.com.my.                 900     IN      A      45.60.243.95

;; Query time: 20 msec
;; SERVER: 127.0.0.53#53(127.0.0.53)
;; WHEN: Wed Dec 04 23:47:01 UTC 2024
;; MSG SIZE rcvd: 70
```


b) yahoo.com
dig yahoo.com

```
osboxes@osboxes:~$ dig yahoo.com

; <<>> DiG 9.16.1-Ubuntu <<>> yahoo.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 23299
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 65494
;; QUESTION SECTION:
;yahoo.com.                IN      A

;; ANSWER SECTION:
yahoo.com.                47      IN      A      74.6.231.21
yahoo.com.                47      IN      A      98.137.11.163
yahoo.com.                47      IN      A      98.137.11.164
yahoo.com.                47      IN      A      74.6.143.25
yahoo.com.                47      IN      A      74.6.231.20
yahoo.com.                47      IN      A      74.6.143.26

;; Query time: 16 msec
;; SERVER: 127.0.0.53#53(127.0.0.53)
;; WHEN: Wed Dec 04 23:47:27 UTC 2024
;; MSG SIZE rcvd: 134
```

c) google.com : Use ANY to return all of the results of the query
dig google.com ANY

```
osboxes@osboxes:~$ dig google.com ANY

; <<>> DiG 9.16.1-Ubuntu <<>> google.com ANY
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 63513
;; flags: qr rd ra; QUERY: 1, ANSWER: 7, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 65494
;; QUESTION SECTION:
;google.com.              IN      ANY

;; ANSWER SECTION:
google.com.              93      IN      NS      ns1.google.com.
google.com.              93      IN      NS      ns4.google.com.
google.com.              93      IN      NS      ns3.google.com.
google.com.              93      IN      NS      ns2.google.com.
google.com.              199     IN      AAAA    2404:6800:4001:810::200e
google.com.              590     IN      TYPE65  \# 13 00010000010000602683202683
3
google.com.              117     IN      A      216.58.199.238

;; Query time: 15 msec
;; SERVER: 127.0.0.53#53(127.0.0.53)
;; WHEN: Wed Dec 04 23:48:28 UTC 2024
;; MSG SIZE rcvd: 180
```

d) mmu.edu.my : Use +trace option, lists each different server the query goes through to its final destination. Use this command option to identify the IP address where traffic is dropping.

dig mmu.edu.my +trace

```
Files s@osboxes:~$ dig mmu.edu.my +trace

; <<>> DiG 9.16.1-Ubuntu <<>> mmu.edu.my +trace
;; global options: +cmd
.                88181    IN      NS      i.root-servers.net.
.                88181    IN      NS      f.root-servers.net.
.                88181    IN      NS      a.root-servers.net.
.                88181    IN      NS      d.root-servers.net.
.                88181    IN      NS      b.root-servers.net.
.                88181    IN      NS      e.root-servers.net.
.                88181    IN      NS      l.root-servers.net.
.                88181    IN      NS      g.root-servers.net.
.                88181    IN      NS      c.root-servers.net.
.                88181    IN      NS      j.root-servers.net.
.                88181    IN      NS      h.root-servers.net.
.                88181    IN      NS      k.root-servers.net.
.                88181    IN      NS      m.root-servers.net.
;; Received 262 bytes from 127.0.0.53#53(127.0.0.53) in 19 ms

my.              172800   IN      NS      a.nic.my.
my.              172800   IN      NS      a.mynic.centralnic-dns.com.
my.              172800   IN      NS      b.mynic.centralnic-dns.com.
my.              172800   IN      NS      c.mynic.centralnic-dns.com.
my.              172800   IN      NS      d.mynic.centralnic-dns.com.
my.              172800   IN      NS      a1.nic.my.
my.              172800   IN      NS      ns01.trs-dns.com.
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Dec 4 23:49

osboxes@osboxes: ~
.      88181 IN    NS    c.root-servers.net.
.      88181 IN    NS    j.root-servers.net.
.      88181 IN    NS    h.root-servers.net.
.      88181 IN    NS    k.root-servers.net.
.      88181 IN    NS    m.root-servers.net.
;; Received 262 bytes from 127.0.0.53#53(127.0.0.53) in 19 ms

my.    172800 IN    NS    a.nic.my.
my.    172800 IN    NS    a.mynic.centralnic-dns.com.
my.    172800 IN    NS    b.mynic.centralnic-dns.com.
my.    172800 IN    NS    c.mynic.centralnic-dns.com.
my.    172800 IN    NS    d.mynic.centralnic-dns.com.
my.    172800 IN    NS    a1.nic.my.
my.    172800 IN    NS    ns01.trs-dns.com.
my.    172800 IN    NS    ns01.trs-dns.net.
my.    86400 IN    DS    174 8 2 FB9E2F10F50A09E3614A9E6
A2C76C1AE7554711E5242B7F516A8078D 86ED87B9
my.    86400 IN    DS    35481 8 2 AE555A40F6BDB488EB025
822A71C91C94E1F339FDD71E0386561C3C4 B7D2B198
my.    86400 IN    RRSIG DS 8 1 86400 20241217220000 202
41204210000 61050 . iOGjQ07w5FitcMYkRZG0DkVo32n8sJwUqoJG4FJr12t5r102DJIXe6Rj cO
g4oZ4LqIYuPNHKAAt3DyMUqEiE0qV8BXLwFAEgaxyszd/L36RD2WZL xS+fUP/kPcxg0aRj1QhIW4PF
sudHm09bd+p6PIKXdBNTId507Td+Na64 rupF1QsBWiu21oxjT4JvMYssxBfFynb5pWaL+6GTo78bWz
3Vf0NwrXGy JEDNoVzM1RgM9KoJh6RJZG+t8VGcsIzrcktPFbLn+ieU0a2KCLSz2DQm 0rwbpsHgL67
HiSgnyp9A3xSvU0HwyjRVLz02T6bs930+9YZTCR+Y1QV To9FFw==
;; Received 900 bytes from 193.0.14.129#53(k.root-servers.net) in 95 ms

edu.my. 86400 IN    NS    ns01.trs-dns.net.
edu.my. 86400 IN    NS    c.mynic.centralnic-dns.com.
```

```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Dec 4 23:50

osboxes@osboxes: ~
edu.my. 86400 IN    NS    ns01.trs-dns.net.
edu.my. 86400 IN    NS    c.mynic.centralnic-dns.com.
edu.my. 86400 IN    NS    a.mynic.centralnic-dns.com.
edu.my. 86400 IN    NS    d.mynic.centralnic-dns.com.
edu.my. 86400 IN    NS    b.mynic.centralnic-dns.com.
edu.my. 86400 IN    NS    b.nic.my.
edu.my. 86400 IN    NS    b1.nic.my.
edu.my. 86400 IN    NS    ns01.trs-dns.com.
edu.my. 86400 IN    DS    57063 8 2 B97039E17079D9EB17FD1
A3408A19218F205C86270D02236270E62B0 1E56187F
e Terminal 86400 IN    DS    23530 8 2 045D1686294FB1295C93B
20c4089z570AC7F1996A7EFD68C2E303535 826CE7BB
edu.my. 86400 IN    RRSIG DS 8 2 86400 20241225201932 202
41125110939 17739 my. RZY9IXMEroSv7bNVbDD2hKf9ltG8kj0SEc3aXxrBBw4wVrW6xE0kLB06
/ZqrqteIDzONcKy+xx00+l8tcS3wKZYN1UYafBHSe+XuoWH+/oDiNHwL xZ7bobnUco2NxucWe34f/Q
ILsAZUuU9NaWhTa8P01rxwb4V5QYe80vLc FUY=
;; Received 528 bytes from 202.171.47.204#53(a1.nic.my) in 0 ms

mmu.edu.my. 86400 IN    NS    ext-ns-3.mmu.edu.my.
mmu.edu.my. 86400 IN    NS    ext-ns-1.mmu.edu.my.
mmu.edu.my. 86400 IN    NS    ext-ns-2.mmu.edu.my.
mmu.edu.my. 86400 IN    NS    ext-ns-4.mmu.edu.my.
a9qfp49darpamfpllf3ov6jl6n3pmh28.edu.my. 3600 IN NSEC3 1 1 1 - AI4BPOLNTCLH3T90
L7GV43E6B2AVQVFD NS SOA RRSIG DNSKEY NSEC3PARAM
a9qfp49darpamfpllf3ov6jl6n3pmh28.edu.my. 3600 IN RRSIG NSEC3 8 3 3600 202412201
73015 20241120074030 61146 edu.my. YQNp0sZGD/Yjl9RJGfpi0vbrWtsJ3Zveybmqi1WMEppW
mq+Tjq4h1dmM d3pHRA6Sssr6DS01JBHUu6qyRqQEuIXNHe71dXtbqUvZfhzZUa2v0Rcd bQ34rK+Kl
NvpX2u16GP/4lJaxjVeef1etHBRQWwMl4IkWkt6QKf58gIq ZxI=
uj0mudfu1etjmbv17hju3c8geip0qhbq.edu.my. 3600 IN NSEC3 1 1 1 - 03FVF8EEU28957G7
```

```

ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Dec 4 23:50
osboxes@osboxes: ~
ILsAZUu9NaWhTa8P01rxwb4V5QYe80vLc FUY=
;; Received 528 bytes from 202.171.47.204#53(a1.nic.my) in 0 ms

mmu.edu.my.      86400    IN       NS       ext-ns-3.mmu.edu.my.
mmu.edu.my.      86400    IN       NS       ext-ns-1.mmu.edu.my.
mmu.edu.my.      86400    IN       NS       ext-ns-2.mmu.edu.my.
mmu.edu.my.      86400    IN       NS       ext-ns-4.mmu.edu.my.
a9qfp49darpamfpll3ov6jl6n3pmh28.edu.my. 3600 IN NSEC3 1 1 1 - AI4BPOLNTCLH3T90
L7GV43E6B2AVQVFD NS SOA RRSIG DNSKEY NSEC3PARAM
a9qfp49darpamfpll3ov6jl6n3pmh28.edu.my. 3600 IN RRSIG NSEC3 8 3 3600 202412201
73015 20241120074030 61146 edu.my. YQNp0sZGD/Yjl9RJGfpi0vbrWtsJ3Zveybmqi1WMEppW
mq+Tjq4h1dmM d3pHRA6Sssr6DSD1JBHUU6qyRqQEuiXNHe71dXtbqUvZfhzZUa2v0Rcd bQ34rK+Kl
NvpX2u16GP/4lJaxjVeef1etHBRQWwL4IkWkT6QKf58gIq ZxI=
uj0mudfu1etjmbv17hju3c8geip0qhbq.edu.my. 3600 IN NSEC3 1 1 1 - 03FVF8EEU28957G7
530VS0881433DLMS NS DS RRSIG
uj0mudfu1etjmbv17hju3c8geip0qhbq.edu.my. 3600 IN RRSIG NSEC3 8 3 3600 202412200
05544 20241120074030 61146 edu.my. rrQVetLcCEm6RRz3XaqWcFFGDEAy5FQZnKgkGxLP50f
MNK0kb73qyvz QKVPgh0W8Ht2mTXkeKle+hQfL+ZBpjL29idw0UbPpXRJoaECTIM8ee06 F1o0kMuWz
nC25z0fxWCM4GoTMM85gC7vmAxG24omjgQEvMWL6Q189oa3 WnY=
;; Received 714 bytes from 202.171.47.205#53(b1.nic.my) in 11 ms

mmu.edu.my.      3600     IN       A        45.60.15.220
mmu.edu.my.      3600     IN       NS       ext-ns-1.mmu.edu.my.
mmu.edu.my.      3600     IN       NS       ext-ns-2.mmu.edu.my.
mmu.edu.my.      3600     IN       NS       ext-ns-3.mmu.edu.my.
mmu.edu.my.      3600     IN       NS       ext-ns-4.mmu.edu.my.
;; Received 211 bytes from 203.106.62.231#53(ext-ns-1.mmu.edu.my) in 15 ms

osboxes@osboxes:~$

```

8. Not all sites run their own DNS servers and not all sites need their own DNS servers! In sufficiently small sites with no Internet connectivity, it is reasonable for each host to keep its own copy of a table matching all of the hostnames on the local network with their corresponding IP addresses. In most Linux and UNIX systems, this table is stored in the `/etc/hosts` file. The `/etc/hosts` file keeps its information in a simple tabular format in a plain-text file. The IP address is in the first column, and all the related hostnames are in the second column. Show the `/etc/hosts` in your Linux server.

```
cat /etc/hosts
```

```

osboxes@osboxes:~$ cat /etc/hosts
127.0.0.1 localhost
127.0.1.1 osboxes

# The following lines are desirable for IPv6 capable hosts
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters

```